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Simulation of partial melting material

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Simulation of partial melting material

by
Chenyu Tian

Dissertation

Presented to the Faculty of the Graduate School of
The University of Texas at Austin
in Partial Fulfillment
of the Requirements
for the Degree of

Doctor of Philosophy

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Dedication

To all future Longhorn graduate students.

Epigraph

What starts here changes the world.

—University of Texas at Austin

Acknowledgments

Thanks to the previous authors of this template, and to the Graduate School for providing it.

Preface

Readers, this document represents a great deal of time and effort. Enjoy.
Cheers, Chenyu Tian.

Abstract

Simulation of partial melting material

Chenyu Tian, PhD
The University of Texas at Austin, 2025

SUPERVISORS: Todd Arbogast, Marc Hesse

Your abstract goes here. Do not use `\abstract`!

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Chapter 1: Introduction

Please be sure to refer to the Graduate School's format guidelines as they may have changed. This template was last updated April 2023.

As of this writing (April 2023) there are several required sections for ETDs, with several optional sections. All required sections and optional sections are supported by this template. To make changes to most formatting options, you will need to update `styles/utformat.sty`. Some aspects of the formatting depend on which kind of document you are producing (dissertation, report, thesis, treatise), so be sure to update the appropriate section if relevant.

1.1 Important Note

Please be sure you have selected the appropriate type of document by using the `\dissertation`, `\mastersthesis`, `\mastersreport`, or `\treatise` commands in your main `.tex` file (these are correct in the provided template for each main document option). Be sure to update the definitions for `\degree` and `\degreeabbr` in your main `.tex` file if appropriate.

1.2 Status of this Template

Below you will find some information on the current status of this template.

1.2.1 Limitations

If you have one appendix, use command `\appendix` in `structural/body.tex`. If you have more than one appendix, use command `\appendices` instead.

Currently, this template does not support dual degrees. In this case, you may find it easiest to edit the definition of `\titlepage` in `styles/utformat.sty`

to say “Degrees” rather than “Degree.” It will be easiest to replace the line that includes `\d@gree` and add additional lines for each degree. See the Graduate School’s formatting document for more information on what the title page should look like for dual degrees.

This template will automatically set the graduation month and year based on when you compile it. If needed, this can be changed at `\graduation@month` in `styles/utformat.sty`.

1.2.2 Formatting Choices

While the current Graduate School formatting requirements are quite flexible, several choices have been made in the development of this template and are detailed below:

- The bibliography section has been titled “Works Cited.” To change this, update the definition of `thebibliography` in `styles/utformat.sty`.
- Figures, tables, etc. are double-numbered, meaning that they reflect the chapter (e.g., Figure 2.1 is the first figure in Chapter 2). To change this, use `\counterwithout{figure}{chapter}` and similar commands – see the commented section near the top of `styles/utformat.tex`.
- The Table of Contents will include chapters, sections and subsections (not sub-subsections). To change this, you will need to update `tocdepth`. This is set in `structural/preamble.tex`.
- Within chapters, sub-subsections will be numbered (e.g., §1.2.3.4). To change this, update `secnumdepth`. This is set in `structural/preamble.tex`.
- See §2 for examples of how citations are currently formatted. This template uses the `natbib` package for citations, which can be customized.

- Text is 1.5-spaced. To change, instead of `\oneandahalfspacing`, you may use `\doublespacing` in `structural/preamble.tex`.
- Block quotes are single-spaced. To change, instead of `\singlespacequote`, use `\doublespacequote` in `styles/utformat.sty`.
- Numbering for theorems and similar environments is set at the chapter level, but can be set in `structural/preamble.tex`.

1.3 Example Section Name

Example section text.

1.3.1 Example Subsection

Example subsection text.

1.3.1.1 Example Subsubsection

Example sub-subsection text.

Chapter 2: Main Text Longer Title¹

Abstract

If your chapter requires an abstract, use this command, NOT `abstract`.

Main text. In-text citation ? or parenthetical citation (?).

2.1 Section Title

Figures can be included as in Figure 2.1. Additional figures will be numbered as Figure 2.2.

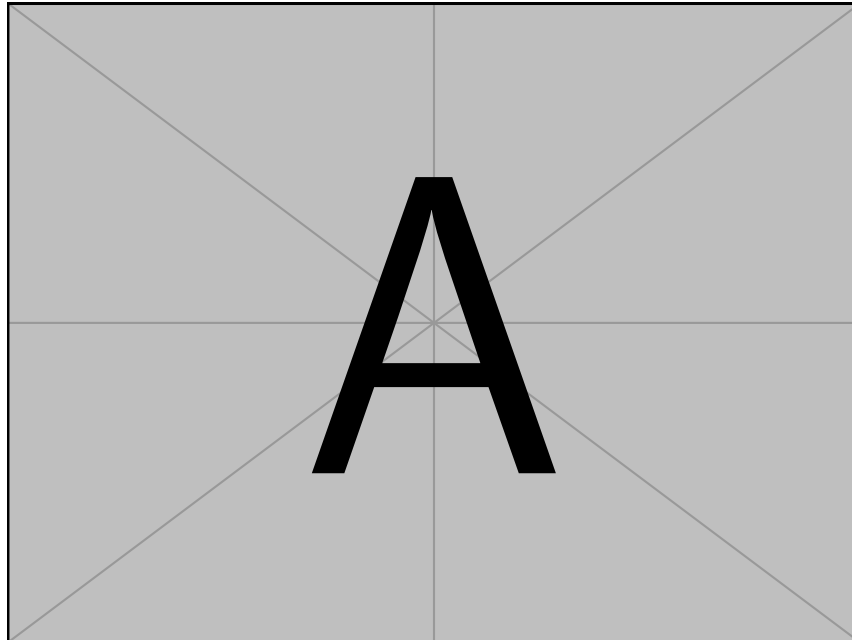


Figure 2.1: Example image

Tables can be included as in Table 2.1.

¹Including a footnote.

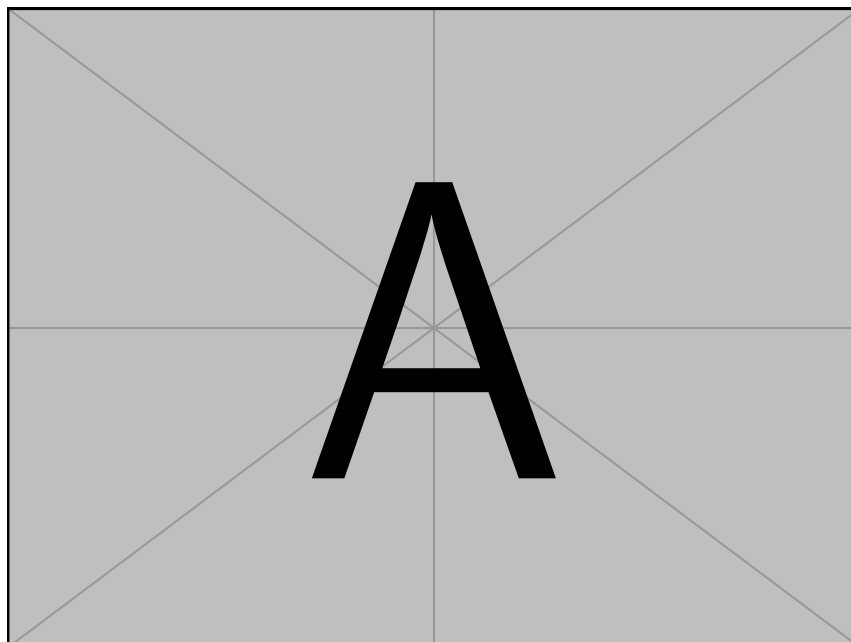


Figure 2.2: Example image

a	b
c	d

Table 2.1: Longer explanation of the 2x2 table shown above.

Illustrations can be included as in Illustration 2.1.

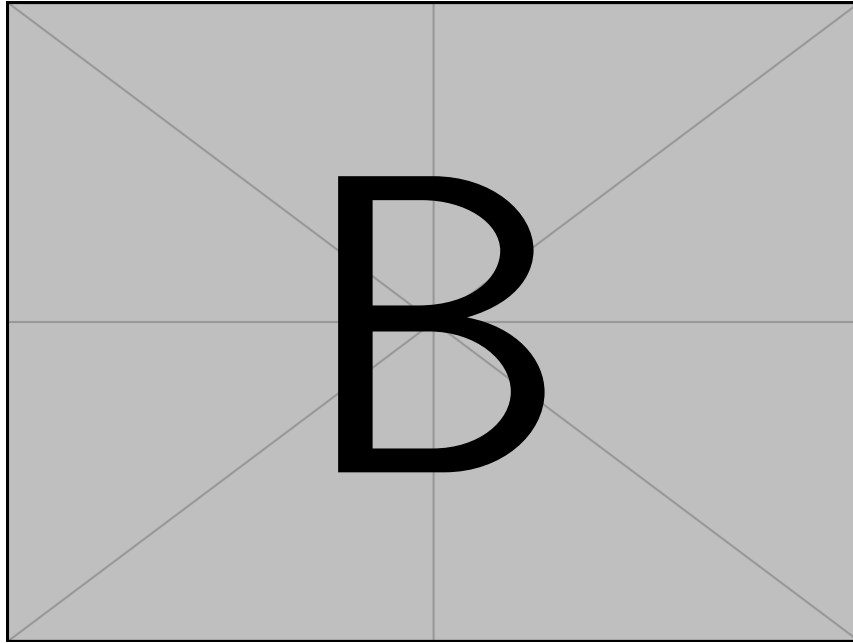
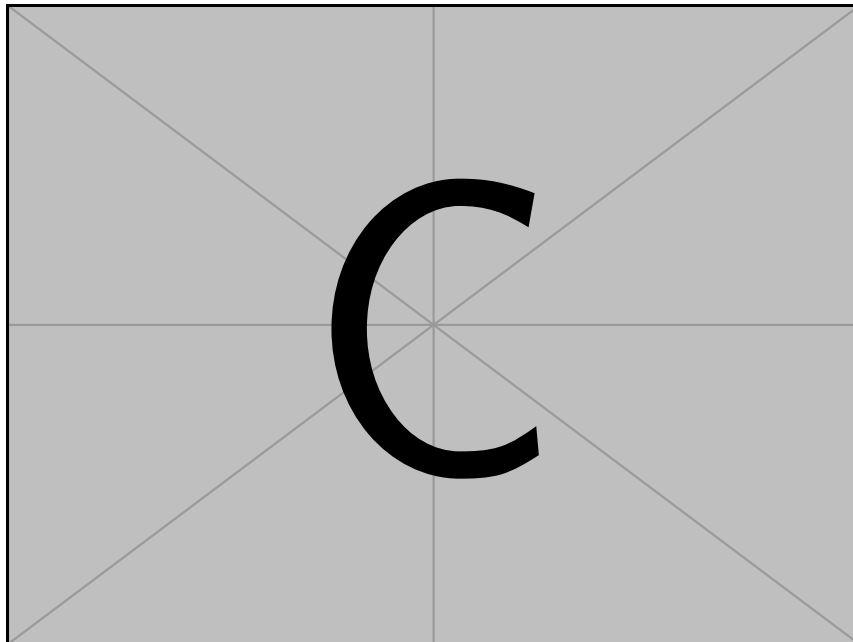


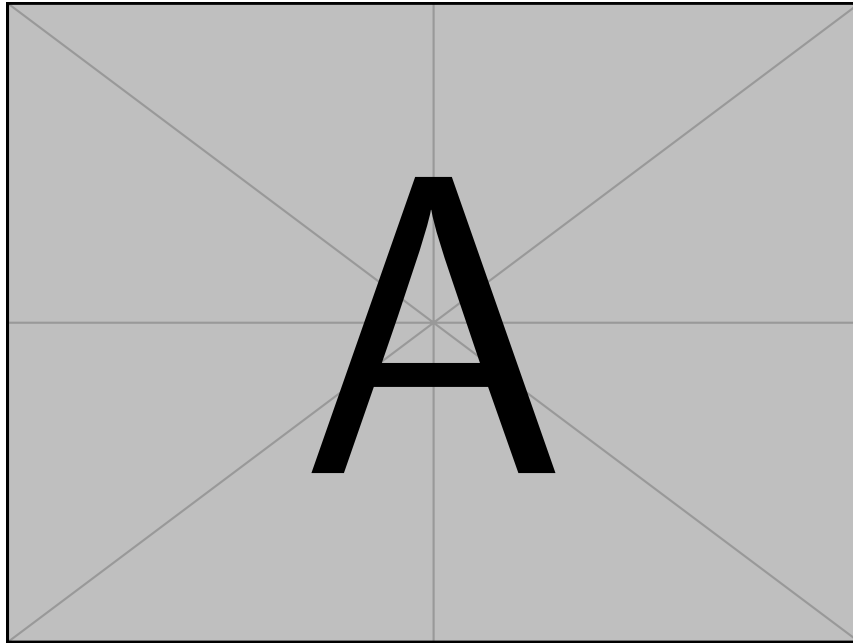
Illustration 2.1: Longer explanation of the illustration shown above.

Maps can be included as in Map 2.1.



Map 2.1: Longer explanation of the map shown above.

Slides can be included as in Slide 2.1.



Slide 2.1: Longer explanation of the slide shown above.

Block quotes can be included as well, using the `quotation` package, as can be seen in the following quotation.

The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.

Theorems can be added as Theorem 2.1, along with definitions, lemmas, etc.

Theorem 2.1. *Theorem here.*

Chapter 3: Governing Equations in Mantle Dynamics

3.1 Mechanics of Mantle Dynamics

Fluid dynamics of the mixture of two phase flow has been extensively discussed. I will discuss the three governing equations of mass and momentum referring to the similar discussion in Drew and Segel (1971), McKenzie (1984) and Katz (2022). Considering a conservation equation for given volumetric variable γ in a representative volume element V .

$$\frac{d}{dt} \int_V \gamma d\mathbf{v} = - \int_{\partial V} \mathbf{f} \cdot d\sigma + \int_V S d\mathbf{v} \quad (3.1)$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V . The minus sign indicates that the positive directions aligns with normal unit vector pointing outwards. S , source/sink term, represents the volumetric production rate of γ . We can apply Gauss's theorem to obtain differential form,

$$\frac{\partial \gamma}{\partial t} + \nabla \cdot \mathbf{f} = S \quad (3.2)$$

We also define the quantity of phase fraction rigorously with the help of an indicator function admitting that multiple phase cannot coexist at the same spatial point, which are regarded as immiscible phases. We define indicator function,

$$i(\mathbf{x}) = \begin{cases} 1 & \text{phase}_i \\ 0 & \text{not phase}_i \end{cases} \quad (3.3)$$

We then define macroscopic phase fraction function:

$$\phi_i = \frac{1}{\delta \mathbf{v}} \int_{\delta \mathbf{v}} i(\mathbf{x}) d\mathbf{v} \quad (3.4)$$

In the beginning of my discussion, I will discuss two immiscible phases liquid magma and solid mantle. Phase fraction of liquid magma is labelled as ϕ_l and phase fraction of solid mantle is labelled as ϕ_s , where $\phi_l + \phi_s = 1$ is always attained.

Substituting bulk density $\bar{\rho} = \phi_l \rho_l + \phi_s \rho_s$ for generalized volumetric variable γ in equation 3.1, I can obtain conservation law for bulk mass,

$$\frac{\partial \bar{\rho}}{\partial t} + \nabla \cdot \bar{\rho} \mathbf{v} = 0 \quad (3.5)$$

with the help of Boussinesq approximation ignoring compressibility of each individual phase, and extend Boussinesq approximation further applying to density difference between each phase except body force term. I can write continuous function for two phase mixture simply as,

$$\nabla \cdot \bar{\mathbf{v}} = 0 \quad (3.6)$$

Substituting phase averaged momentum $\bar{\rho} \mathbf{v}$, I then write conservation of momentum with phase averaged term,

$$\frac{\partial \bar{\rho} \mathbf{v}}{\partial t} + \nabla \cdot \overline{\rho \mathbf{v} \otimes \mathbf{v}} = \bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{P} \quad (3.7)$$

where $\boldsymbol{\tau}$ represents deviatoric stress, Ignoring change of inertia term due to near zero Reynolds number in magma and mantle,

$$\bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \bar{P} = 0 \quad (3.8)$$

Insert more details later!

Combining equations, I finally arrive with the full flow mechanical system,

$$\phi_f (\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l} \nabla (p_l - p_s) = 0, \quad (3.9)$$

$$\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) = 0, \quad (3.10)$$

$$\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) = (\rho_s - \rho_l) \mathbf{g}, \quad (3.11)$$

$$\mu_s \nabla \cdot \mathbf{v}_s - \phi_f (p_l - p_s) = 0, \quad (3.12)$$

where the deviatoric stress of the mixture is

$$\tau(\mathbf{v}_s) = 2\mu_s\phi_s(\mathcal{D}\mathbf{v}_s - \frac{1}{3}\nabla \cdot \mathbf{v}_s\mathbf{I}) \quad (3.13)$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla\mathbf{v}_s + \nabla\mathbf{v}_s^T)$ is the symmetric gradient. Moreover, the relative permeability is $k_{\phi_l} = k_0\phi_l^{2+2\Theta}$, where $\Theta \in [0, 1/2]$. μ_s and μ_l are dynamic viscosities for solid and liquid correspondingly.

3.2 Conservation of species and energy

3.3 Eutectic Phase Behavior

Embracing the fact that Mid-ocean ridge forms at the place where temperature is generally lower, eutectic melting, where melting temperature is significantly lowered with the presence of the secondary component, is a good model for modelling melting behavior at certain location. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution. On a typical eutectic phase behavior diagram, we can locate any phase situation using temperature and composition information. I follow phase behavior details described in Marc A. Hesse and Oza (2022). Some adjustments are made for our mantle system.

- Fluid formed by eutectic melting has a constant composition of opx denoted as X_e . The eutectic composition can be related to pressure and denoted as $X_e(P)$.
- Melting temperature relates to pressure. Decompression melting is accounted for in our model by relating eutectic melting temperature with pressure change.
- Eutectic melting is of the most interest. The temperature where eutectic melting happens has much lower temperature than actual melting temperature.

3.4 Rescaling and Nondimensionalization

Appendix A: Appendix Name

Text here.

A.1 Appendix section

Text here.

A.1.1 Appendix section

Text here.

A.1.1.1 Appendix section

Text here.

Appendix B: Another Appendix

Glossary

Chenyu Tian A hard-working graduate student, nearing the end of this chapter of your education!

Doctor of Philosophy Chenyu Tian is about to earn this. Congratulations!

Works Cited

Donald Drew and Lee Segel. Averaged equations for two-phase flows. *Studies in Applied Mathematics*, page 3, 1971.

Richard F. Katz. *The Dynamics of Partially Molten Rock*. Princeton University Press, 1st edition, 2022. ISBN 9780691232645.

Steve D. Vance Marc A. Hesse, Jacob S. Jordan and Apurva V. Oza. Supporting information for "downward oxidant transport through europa's ice shell by density-driven brine percolation". *Geophysical Research Letters*, 2022.

Dan Mckenzie. The generation and compaction of partially molten rock. *Journal of Petrology*, 25:713–765, 1984.

Vita

Chenyu Tian is going to graduate soon!

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This dissertation was typeset with $\text{\LaTeX}^\dagger$ by the author.

[†] $\text{\LaTeX}$ is a document preparation system developed by Leslie Lamport as a special version of Donald Knuth's $\text{\TeX}$ Program.